

MATH 114 HW 5

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Question 1

- a. Note that $B(t) - B(s) \sim N(0, t - s)$ and $B(0) = 0$

$$\begin{aligned} B(8) &= B(8) - B(0) \sim N(0, 8 - 0) \\ \mathbb{E}(B(8)) &= \mathbb{E}(B(8) - B(0)) \\ &= 0 \\ \text{Var}(B(8)) &= \text{Var}(B(8) - B(0)) \\ &= 8 \\ B(8) - B(5) &\sim N(0, 8 - 5) \\ &\sim N(0, 3) \\ \text{Var}(B(8) - B(5)) &= 3 \end{aligned}$$

- b. Note that $\Phi(z) = \mathbb{P}(Z < z)$ for $Z \sim N(0, 1)$

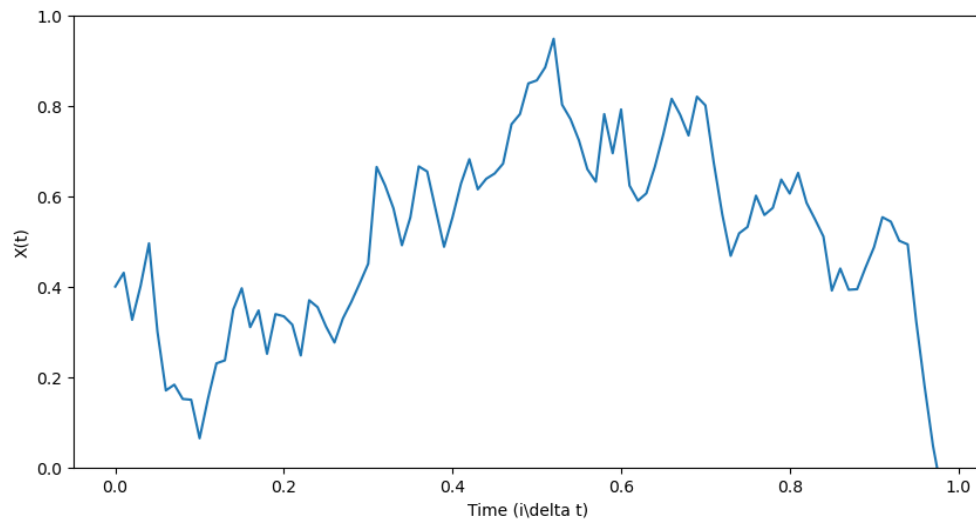
$$\begin{aligned} \mathbb{P}(B(4) \leq 3 \mid B(1) = 1) &= \mathbb{P}(B(4) - B(1) \leq 3 - 1 \mid B(1) = 1) \\ &= \mathbb{P}(B(4) - B(1) \leq 2 \mid B(1) = 1) \end{aligned}$$

Since $B(4) - B(1) \sim N(0, 3)$, we have

$$\begin{aligned} \text{Var}Z &= 1 \\ 3\text{Var}Z &= 3 \\ \text{Var}(\sqrt{3}Z) &= 3 \\ \text{Var}(\sqrt{3}Z) &= \text{Var}(B(4) - B(1)) \end{aligned}$$

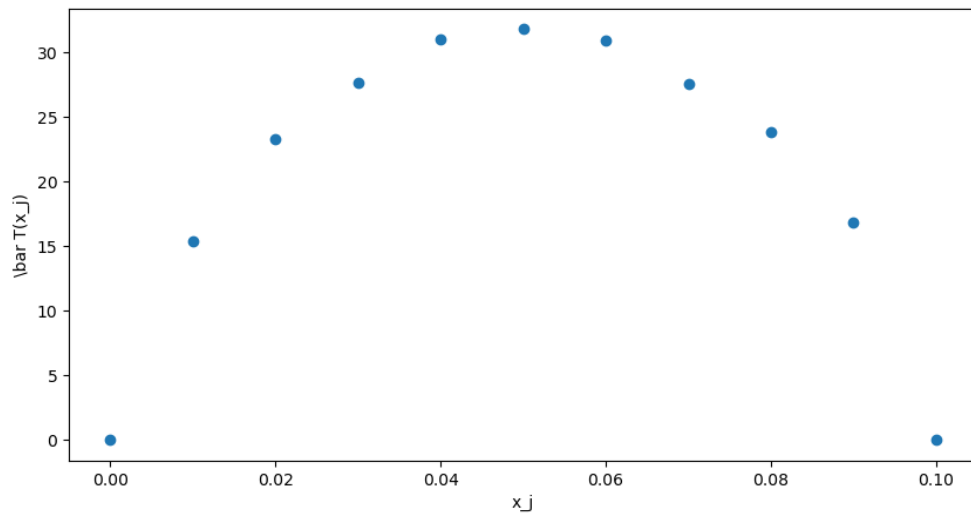
$$\begin{aligned}\mathbb{P}(\sqrt{3}Z \leq 2) &= \mathbb{P}(Z \leq \frac{2}{\sqrt{3}}) \\ &= \Phi\left(\frac{2}{\sqrt{3}}\right)\end{aligned}$$

Question 2



a.

b. The estimator $\bar{T}(0.4) = 30.985$



c.

Question 3

a. Given $X(0) = 1$

$$\begin{aligned}dX(t) &= -X(t)dt + \sqrt{2}dB(t) \\ \mathbb{E}(dX(t)) &= \mathbb{E}(-X(t)dt) + \mathbb{E}(\sqrt{2}dB(t)) \\ d\mathbb{E}(X(t)) &= -\mathbb{E}(X(t))dt + 0 \\ \frac{d}{dt}\mathbb{E}(X(t)) &= -\mathbb{E}(X(t)) \\ \mathbb{E}(X(t)) &= Ce^{-t}\end{aligned}$$

Plug in the initial condition $X(0) = 1$

$$\begin{aligned}\mathbb{E}(X(0)) &= Ce^{-0} \\ 1 &= C\end{aligned}$$

Therefore we have

$$\begin{aligned}\mathbb{E}(X(t)) &= e^{-t} \\ \mathbb{E}(X(1)) &= e^{-1}\end{aligned}$$

b. i. Note that $X \sim N(\mu, \sigma^2)$ if $X = \sigma Z + \mu$

$$\begin{aligned}X_1 &= X_0 - X_0\delta t + \sqrt{2\delta t}Z_0 \\ &= 1 - \delta t + \sqrt{2\delta t}Z_0 \\ X_1 &\sim N(1 - \delta t, 2\delta t)\end{aligned}$$

ii. If X_i is normally distributed

$$X_{i+1} = (1 - \delta t)X_i + (\sqrt{2\delta t})Z_i$$

Sum of normals are normals, therefore, X_{i+1} is normally distributed

iii. Find $\mathbb{E}(X_{i+1})$

$$\begin{aligned}\mathbb{E}(X_{i+1}) &= (1 - \delta t)\mathbb{E}(X_i) + \sqrt{2\delta t}\mathbb{E}Z_i \\ &= (1 - \delta t)\mathbb{E}(X_i) \quad \text{since } \mathbb{E}Z_i = 0\end{aligned}$$

Find $\mathbb{E}(X_i)$

$$\begin{aligned}
\mathbb{E}(X_i) &= (1 - \delta t)\mathbb{E}(X_{i-1}) \\
&= (1 - \delta t)^2\mathbb{E}(X_{i-2}) \\
&\vdots \\
&= (1 - \delta t)^i\mathbb{E}(X_0) \\
&= (1 - \delta t)^i \quad \text{since } \mathbb{E}(X_0) = 1
\end{aligned}$$

iv. Find $\text{Var}(X_{i+1})$

$$\text{Var}(X_{i+1}) = \text{Var}((1 - \delta t)X_i + \sqrt{2\delta t}Z_i)$$

Note that $\text{Var}(aX + bY) = a^2\text{Var}X + b^2\text{Var}Y + 2ab\text{Cov}(X, Y)$, but since X, Y independent in this case, $\text{Cov}(X, Y) = 0$

$$\begin{aligned}
\text{Var}(X_{i+1}) &= (1 - \delta t)^2\text{Var}(X_i) + 2\delta t\text{Var}(Z_i) \\
&= (1 - \delta t)^2\text{Var}(X_i) + 2\delta t
\end{aligned}$$

Find $\text{Var}(X_i)$

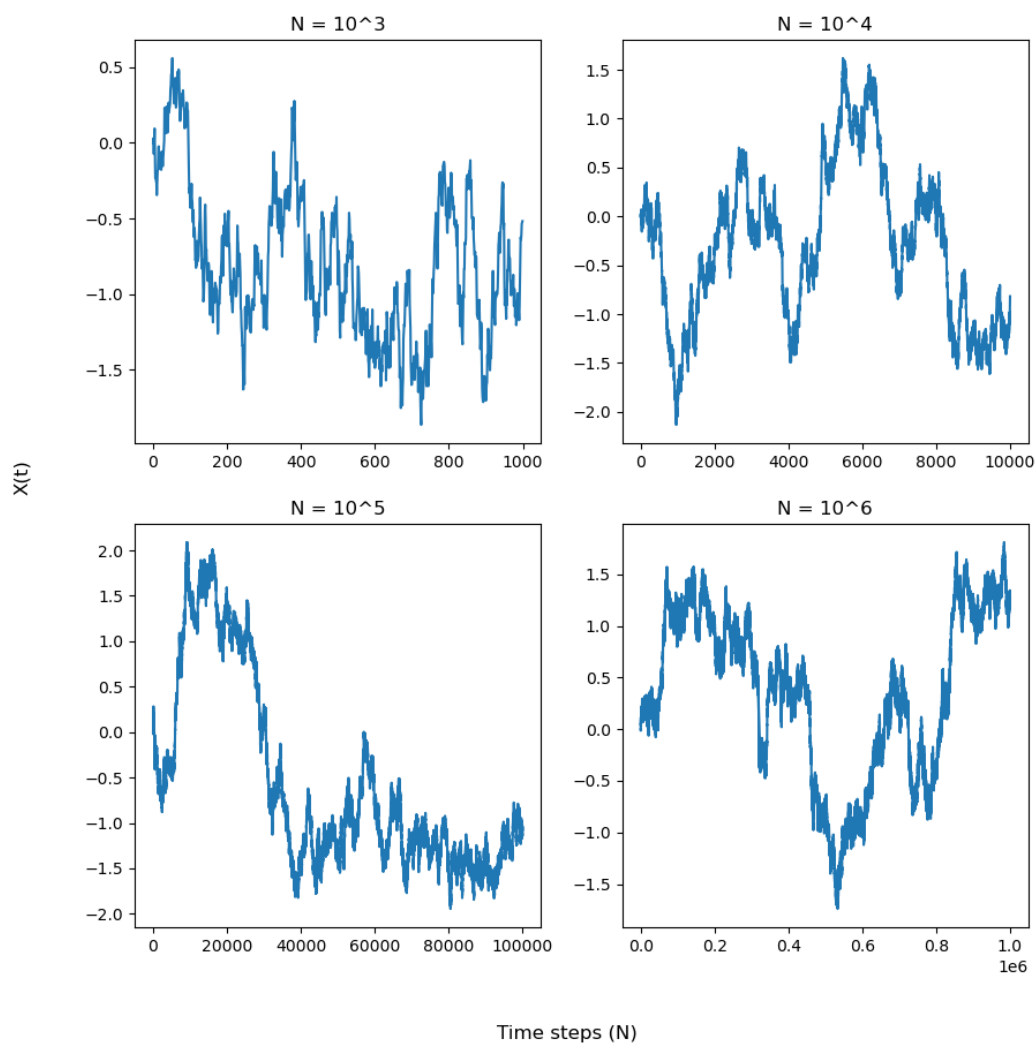
$$\begin{aligned}
\text{Var}(X_i) &= (1 - \delta t)^2(\text{Var}(X_{i-1})) + 2\delta t \\
&= (1 - \delta t)^2((1 - \delta t)^2\text{Var}(X_{i-2}) + 2\delta t) + 2\delta t \\
&= (1 - \delta t)^4\text{Var}(X_{i-2}) + (1 - \delta t)^2 \cdot 2\delta t + 2\delta t \\
&= (1 - \delta t)^4\text{Var}(X_{i-2}) + 2\delta t(1 + (1 - \delta t)^2) \\
&= (1 - \delta t)^4((1 - \delta t)^2\text{Var}(X_{i-3}) + 2\delta t) + 2\delta t(1 + (1 - \delta t)^2) \\
&= (1 - \delta t)^6(\text{Var}(X_{i-3})) + 2\delta t(1 + (1 - \delta t)^2 + (1 - \delta t)^4) \\
&\vdots \\
&= (1 - \delta t)^{2i}\text{Var}(X_0) + 2\delta t \sum_{j=0}^{i-1} (1 - \delta t)^{2j} \\
&= 2\delta t \sum_{j=0}^{i-1} (1 - \delta t)^{2j}
\end{aligned}$$

v. $X_i \sim N\left((1 - \delta t)^i, 2\delta t \sum_{j=0}^{i-1} (1 - \delta t)^{2j}\right)$

Question 4

Define $U'(x) = x(x^2 - 1) = x(x + 1)(x - 1)$

Euler-Maruyama method for SDE with different N



The position of the particles hover around 1 and -1 with some noise, and when the noise bumps it past zero, it falls towards the other side, hovering again around -1 or 1.